

Trident G Theory of General Adaptive Intelligence (v2)

A dynamical characterisation of intelligence as Griffiths Phase bifurcating dynamics

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General adaptive intelligence (G) is the capacity to maintain viable near-critical cognition across challenge, search relational problem spaces without collapsing into rigidity or fragmentation, validate useful structures across variation, and crystallise what survives into reusable Gc that can later be re-entered flexibly under new Gf demand.

Trident-G theory offers a unified architecture for how adaptive intelligence is maintained, expressed, destabilised, trained, validated and crystallised over time.

1. Structural Overview

The G loop is not a simple sequential process. It is a concurrent multi-layer dynamical system in which specific transition events mark qualitative regime changes.

Four layers operate simultaneously, coupled through:

- shared α stability monitoring
- the hippocampal successor representation (SR) map system
- fast and slow Gc construction pathways
- allostatic regulation through the Ψ -band

Each loop iteration is a spiral rather than a cycle. It returns to a representationally richer starting state through the slow Gc gist extraction pathway, which installs cross-loop invariants into cortical representational geometry.

Part I — The Four-Layer G Loop Architecture

2. Layer 1: The Sustained Bifurcating Dynamic

Always running — constitutes the Griffiths Phase regime itself

Layer 1 is the core dynamical engine. It maintains the system in an extended near-critical regime through the competition between entropy-driven exploration and mutual-information-driven constraint.

2.1 FPCN-B Entropy Arm Activation

The FPCN-B entropy arm supports broad associative search across relational map space via the hippocampal intermediate-horizon SR system.

This is not random exploration. It is SR-structured search:

- successor representations are weighted by temporal discounting
- candidate relational configurations are generated
- these configurations are neither purely local nor purely global
- the default mode network contributes autobiographical and semantic depth

This corresponds to the Zhang–Tang entropy maximisation process: the system maintains maximal flexibility across the current state of knowledge.

The entropy arm's reach depends on the richness of the intermediate-horizon SR maps installed by prior loop iterations through the slow Gc pathway.

2.2 FPCN-A Mutual Information Constraint

The FPCN-A mutual-information arm applies simultaneous goal-directed constraint to the relational configurations generated by the entropy arm.

Its functions include:

- weighting candidate configurations by task relevance
- using prediction-error signals from real-world outcomes
- maintaining the current objective function in working memory
- constraining search without prematurely closing it

This corresponds to the Zhang–Tang mutual information constraint: updates are filtered for task-relevant dependency between representations and outcomes.

The precision of this constraint arm depends on the quality of fast Gc compilations available as structured building blocks from prior loop iterations.

2.3 Griffiths Phase Regulation

The maintained competition between FPCN-B entropy and FPCN-A mutual information prevents the system from collapsing into either:

- pure exploitation
- pure exploration

This balance is monitored through α stability: whether the system's representational update distribution maintains heavy-tailed, power-law statistics consistent with Griffiths Phase operation.

Regulation rules

When mutual information saturation is detected:

- the update distribution flattens toward thin tails
- the system risks subcritical collapse
- entropy injection is triggered

When entropy excess is detected:

- update distributions become extremely heavy-tailed
- convergence fails
- mutual-information tightening occurs

This regulation keeps the system within the Griffiths Phase regime rather than drifting into subcritical rigidity or supercritical fragmentation.

The GP regulation function is the computational analogue of regional heterogeneity and structural modularity producing an extended critical regime. It prevents any single module's bifurcation resolution from dominating the global dynamic.

3. Layer 2: The SR Tuning Dynamic

Operating within the GP regime on the currently active relational map

Layer 2 describes how the active SR map is tuned, extended, and coupled across timescales.

3.1 Horizontal SR Tuning

Within the currently active relational map, the system tunes its successor representation across surface variation.

Examples of variation include:

- different wrappers
- different feature instantiations
- different contextual embeddings

This is the interior of a power-law learning segment: efficient exploitation of the current relational structure generating smooth performance gains.

The key variable is the SR temporal horizon.

Horizontal tuning deepens the map's predictive accuracy at the current horizon without yet extending it.

Characteristic signature

- monotonically improving performance
 - decreasing update-magnitude variance
 - progressive mutual-information constraint
 - no full saturation yet
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3.2 Temporal Horizon Probing

Simultaneously, the system probes whether extending the SR temporal horizon reveals deeper relational structure.

This means asking whether the current map's predictions improve if the discount factor is relaxed to include more distal state transitions.

This is not deliberate strategy. It is an automatic property of the hippocampal SR system.

When longer-horizon predictions add explanatory value:

- the active SR map extends its horizon
- deeper structural features become visible
- a bifurcation response may begin before performance fully plateaus

This probe allows the loop to self-initiate a bifurcation response before the asymptote is completely reached.

3.3 Cross-Scale Coupling

Three timescales are coupled rather than sequential:

Timescale	Function
Fast subcortical SR automations	Provide compiled Gc building blocks
Intermediate hippocampal SR maps	Recombine structured relational material
Slow prefrontal schemas	Provide boundary conditions and strategic constraints

The coupling runs in both directions:

1. subcortical automations feed upward into hippocampal SR maps
2. hippocampal SR maps feed upward into prefrontal schemas
3. prefrontal schemas feed downward to reshape what the hippocampal SR treats as relevant
4. this reshapes which subcortical automations are recruited

The richness of this cross-scale coupling determines the maximum complexity of relational structures available to the current loop iteration.

- sparse coupling → shallow relational maps
 - rich coupling → deep relational maps
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4. Layer 3: The Active Gc Construction Dynamic

Operating continuously on two distinct pathways

Layer 3 explains how Gf activity becomes Gc structure through fast compilation and slow gist extraction.

4.1 Prediction-Error Gc Pathway — Fast Compilation

When the FPCN-A constraint becomes sufficiently strong, the bifurcating dynamic resolves.

This occurs when a specific relational configuration has been validated across surface variation so that uncertainty about its task relevance is genuinely low.

At this point, the configuration crosses the bifurcation threshold into subcritical SR automation.

This is fast Gc banking.

Features of fast Gc banking

- specific compiled strategies
- documented boundary conditions
- portability signatures
- timescale: deliberate practice, sessions to weeks
- model-free and subcortical-dominant
- post-bifurcation

Once compiled, the structure no longer internally bifurcates, but it remains available to FPCN-B as a structured building block for novel recombination.

4.2 Self-Organising Gc Pathway — Slow Gist Extraction

The slow pathway operates continuously below the threshold of deliberate attention.

The hippocampal system extracts statistical regularities recurring across:

- episodes
- fast Gc compilations

- domains
- contexts
- loop iterations

This gist is progressively installed in cortical representational geometry through consolidation, especially during sleep and rest.

Output

The output is not a compiled strategy. It is a schematic geometry: a restructured representational landscape in which certain relational invariants become primary dimensions.

This makes future entropy-arm search more likely to land on deep portable solutions rather than surface recombinations.

Timescale

- sleep consolidation
- rest intervals
- extended practice
- weeks to months

4.3 Consolidation Interval Dependency

The slow Gc pathway requires genuine consolidation intervals: periods without new prediction-error override.

Without adequate consolidation:

- episodic material accumulates
- schematic geometry is not installed
- the loop loses its spiral property
- each iteration returns to approximately the same representational starting state

In this case, the loop degrades into flat cycling rather than self-enriching spiralling.

Adequate consolidation is therefore not a rest period in a weak sense. It is an active architectural requirement for long-term intelligence development.

Part II — Layer 4: Individual Differences

5. Layer 4: The Individual Differences Layer

Characteristic loop signatures across stable FPCN-A/B balance profiles

The individual differences layer characterises how stable biases in FPCN-A/B balance produce distinctive patterns across the loop's phases and transition events.

Rather than treating individual differences as fixed capacity variations, this layer treats them as characteristic dynamical signatures.

These signatures appear in:

- α stability maintenance
- transition event timing
- Gc pathway activation
- where the loop is most constrained

Five archetypal profiles are identified. Real individuals are blends of these profiles, and profiles can shift across domains and developmental periods.

6. Profile 1: Balanced GP

High Gf, integrated Gc/Gf

6.1 Dynamical Signature

α stability is maintained across all loop phases without characteristic collapse points.

Transition events occur at approximately theoretically optimal timing:

- asymptote detection before complete plateau
- bifurcation response generates genuine higher-order structure
- portability validation cascade completes before fast Gc banking

Both Gc pathways are active and well-coupled:

- fast Gc compilations are genuinely portable
 - slow Gc gist extraction reshapes subsequent search spaces
 - consolidation intervals are used effectively
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6.2 Layer 1 Signature

FPCN-A/B balance is dynamically modulated by task demands.

- entropy arm leads on novel configurations
- mutual-information arm leads on validation and constraint
- neither dominates long enough to collapse the bifurcating dynamic

The product $\mathbb{I} \cdot \beta$ remains approximately stable across loop phases, indicating genuine self-organised criticality rather than externally imposed balance.

6.3 Layer 2 Signature

Horizontal SR tuning is efficient but not premature.

Temporal horizon probing is active and productive. Horizon extensions frequently reveal genuine structural depth, creating self-initiated bifurcation responses before full performance plateau.

Cross-scale coupling is rich:

- subcortical automations
- hippocampal SR maps
- prefrontal schemas

are all mutually coupled and mutually reshaping.

6.4 Layer 3 Signature

Fast Gc compilations are genuinely cross-domain portable.

They survive:

- wrapper variation
- boundary testing

- novel context deployment

Slow Gc gist extraction is productive, installing cross-loop invariants as schematic geometry.

6.5 Training Implication

This profile benefits most from increasing the complexity and novelty of relational structures.

Appropriate challenge level:

- small-world composite problems
- applied scenario training
- top of the vertical stack

Primary risk:

- complacency
 - insufficient novelty
 - over-consolidation by the mutual-information arm without bifurcation response
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7. Profile 2: Mutual Information Dominant

High Gc, constrained Gf

7.1 Dynamical Signature

α stability collapses early in the bifurcation response phase.

The update distribution flattens toward thin tails before genuine higher-order structure is found.

Transition Event 2 is truncated:

- entropy arm activates
- mutual-information arm recaptures the dynamic too early
- existing Gc schema constraints are applied prematurely

- fast Gc banking occurs before full portability validation

This produces wrapper-specific automations that appear robust in familiar contexts but fail under genuine surface variation.

7.2 Layer 1 Signature

FPCN-A activates strongly and quickly on novel problems.

The mutual-information arm applies existing constraint schemas before FPCN-B has completed its search.

The $\mathbb{I} \cdot \beta$ product is unstable toward mutual-information dominance:

- λ rises quickly
- β collapses
- α falls below the optimal heavy-tailed range

The system becomes subcritical during what should be the GP bifurcation response phase.

7.3 Layer 2 Signature

Horizontal SR tuning is highly efficient within familiar relational structures.

This profile often shows steep within-segment power-law performance curves.

However:

- temporal horizon probing is limited
 - horizon extension is blocked by existing schema constraints
 - cross-scale coupling is weighted toward subcortical-to-hippocampal direction
 - existing automations dominate recombination space
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7.4 Layer 3 Signature

Fast Gc compilations accumulate rapidly but lack genuine portability.

Boundary conditions are underspecified and wrapper sensitivity is high.

Slow Gc gist extraction operates on a narrow episodic base, producing refined domain knowledge rather than genuinely cross-domain invariants.

7.5 Training Implication

This profile requires entropy injection at the bifurcation response phase.

Useful training features:

- tasks that resist existing FPCN-A constraint schemas
- non-verbal geometric n-back
- temporal relation n-back
- explicit wrapper variation before consolidation
- boundary case exposure before fast banking

The Donner-Hardy breakpoint should be deliberately extended in training so that the person repeatedly encounters the plateau before consolidation is permitted.

The aim is to build tolerance for the bifurcation response phase rather than permitting premature escape into fast Gc banking.

8. Profile 3: Entropy Dominant

High creativity, constrained consolidation

8.1 Dynamical Signature

α stability collapses in the opposite direction.

The update distribution develops extreme heavy tails without convergence.

Transition Event 3 is delayed or fails:

- many candidate relational configurations are generated
- mutual-information constraint is too weak to select among them
- fast Gc banking is rare
- the bifurcation threshold is rarely crossed

Slow Gc gist extraction is active but poorly anchored because there are insufficient validated fast Gc structures.

8.2 Layer 1 Signature

FPCN-B activates strongly and broadly on novel problems.

The entropy arm generates rich, far-reaching associative search.

However, FPCN-A applies insufficient constraint:

- mutual-information engagement is intermittent or weak
- $\mathbb{I} \cdot \beta$ is unstable toward entropy dominance
- β remains high
- λ fails to rise sufficiently
- α exceeds the optimal heavy-tailed range

This creates supercritical rather than GP-regime dynamics.

8.3 Layer 2 Signature

Horizontal SR tuning is inefficient because entropy dominance prevents stable constraint.

Temporal horizon probing is highly active but undirected.

Horizon extensions are frequent but not filtered for task relevance, producing a broad but shallow SR map.

Cross-scale coupling is weighted toward hippocampal-to-prefrontal flow:

- diverse episodic material accumulates
 - stable subcortical automations are not formed reliably
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8.4 Layer 3 Signature

Fast Gc compilations are rare and often unstable.

Boundary conditions are poorly specified, and performance under pressure is fragile.

Slow Gc gist extraction is rich in episodic diversity but produces schematic geometry that is creative yet unreliable.

8.5 Training Implication

This profile requires mutual-information constraint training.

Critical training priorities:

- FPCN-A goal-relevant constraint
- MFT-M cognitive control at the base of the vertical stack
- structured portability validation cascades
- clear pass/fail criteria
- consolidation discipline

This profile tends to resist consolidation by continuing to explore rather than allowing the slow Gc pathway to install gist as stable schematic geometry.

9. Profile 4: SR Depth Deficient

Limited relational map richness

9.1 Dynamical Signature

α stability is maintained but at a shallow level.

The GP regime is present but operates over a thin relational map. This limits both:

- entropy arm reach
- mutual-information constraint precision

The bifurcating dynamic runs correctly but on an impoverished representational substrate.

Transition Event 3 produces formally correct structures that pass portability validation, but they are not deep because the underlying SR maps lack sufficient relational complexity.

9.2 Layer 1 Signature

FPCN-A/B balance is appropriately maintained but over limited associative territory.

The $\mathbb{I} \cdot \beta$ product is stable, indicating genuine GP dynamics, but the entropy arm's search space is constrained by shallow SR map depth.

The update distribution is heavy-tailed, but the tail only reaches shallow relational territory.

9.3 Layer 2 Signature

Horizontal SR tuning reaches asymptote quickly because the current relational map has limited complexity.

Temporal horizon probing produces limited return because extending the horizon reveals further shallowness rather than deeper structure.

Cross-scale coupling is present, but the building blocks are shallow.

9.4 Layer 3 Signature

Both Gc pathways function correctly but on thin material.

Fast compilations are portable but not powerful.

Slow gist extraction produces genuine invariants that are low-dimensional and do not substantially reshape future search spaces.

9.5 Training Implication

This profile benefits most from the middle levels of the vertical stack:

- relational n-back
- non-verbal geometric relations
- temporal relations

- explicit relational reasoning with broad surface variation

The key diagnostic marker is rapid asymptote across many domains.

The intervention target is increasing structural complexity and temporal depth while preventing semantic Gc retrieval shortcuts.

10. Profile 5: Consolidation Deficient

Active GP, degraded spiral

10.1 Dynamical Signature

Layer 1 and Layer 2 dynamics are approximately Profile 1 quality:

- GP bifurcating dynamic is well maintained
- SR tuning is efficient
- bifurcation response produces genuine higher-order structures

However, the loop fails to spiral.

Each iteration returns to approximately the same representational starting state because slow Gc gist extraction fails to install cross-loop invariants as stable schematic geometry.

Fast Gc compilations accumulate but do not feed productively into the slow Gc pathway.

The loop cycles rather than spirals.

10.2 Layer 1 Signature

GP dynamics are well maintained within each loop iteration.

However, cross-iteration continuity is poor.

The entropy arm's search space at the start of each new iteration is not demonstrably richer than before, indicating slow Gc pathway failure.

10.3 Layer 2 Signature

SR tuning within each iteration is efficient and horizon probing is productive.

The failure occurs at the slow timescale:

- prefrontal schemas are not updated by slow Gc schematic installation
- boundary conditions remain constant across loop iterations
- search spaces do not progressively refine

10.4 Layer 3 Signature

The fast Gc pathway functions correctly.

Compilations are genuinely portable and boundary conditions are well specified.

The slow Gc pathway fails because of one or more of the following:

- insufficient consolidation intervals
- insufficient cross-domain episodic diversity
- impaired cortical consolidation
- narrow training material

The characteristic marker is a library of portable strategies without coherent higher-level schematic integration.

This is expertise without wisdom.

10.5 Training Implication

This profile requires two main interventions.

1. Consolidation interval discipline

Genuine rest periods without prediction-error override are needed so the slow Gc pathway can complete gist extraction.

2. Cross-domain episodic diversity

The slow Gc extractor needs material from genuinely different domains and contexts to find cross-loop invariants.

Useful training:

- small-world composite tasks
- applied scenario training
- top-of-stack integration work
- sleep hygiene
- stress management

Sleep and recovery are not peripheral concerns. They are direct interventions on the slow Gc consolidation mechanism.

Part III — Transition Events

11. Overview

Transition events mark qualitative regime changes across all layers simultaneously.

12. Transition Event 1: Asymptote Detection

The current SR map's performance gains flatten. The power-law learning segment reaches its local ceiling.

This is detected by the GP regulation subsystem as mutual-information saturation:

- prediction-error signal no longer generates sufficient update magnitude
- the update distribution flattens toward thin tails
- the system approaches subcriticality within the current map

Individual differences marker

Profile	Characteristic pattern
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Profile 2	Detects asymptote late, after subcritical collapse
Profile 3	Continues exploring past the plateau
Profile 1	Detects near optimal timing
Profile 4	Detects near optimal timing, but at shallow depth

13. Transition Event 2: GP Bifurcation Response

The FPCN-B entropy arm reactivates at increased intensity.

The hippocampal SR temporal horizon extends beyond its current tuned value.

The default mode network's associative reach expands into domains not previously connected to the current relational structure.

FPCN-A mutual-information constraint relaxes. It does not disappear, but loosens enough to allow novel relational configurations to be evaluated.

α stability signature

- heavy tails extend
- the system moves away from local asymptote
- relational search territory opens

Profile-specific failures

Profile	Failure mode
Profile 2	FPCN-A recaptures the dynamic too early
Profile 3	The event continues indefinitely without convergence

14. Transition Event 3: Higher-Order Structure Emergence

A qualitatively better relational configuration becomes available through entropy-arm search.

The new structure:

- accounts for the current map's successes
- explains its failures as boundary cases
- reveals a more general invariant

In Zhang–Tang terms, this appears as:

- a large update event
- one of the power-law distributed jumps
- rapid mutual-information gain afterwards

Profile 4's limitation occurs here: the new structure is genuine but shallow because SR depth constrains invariant complexity.

15. Transition Event 4: Portability Validation Cascade

The new structure is deployed across:

- wrapper variations
- boundary cases
- genuinely novel contexts

The GP regulation subsystem monitors whether α stability is maintained across these variations.

If α stability is maintained, the structure has genuine cross-domain portability.

If α stability collapses, the structure is a local solution masquerading as a general one, and Transition Event 2 is re-triggered.

Profile-specific failures

Profile	Failure mode
Profile 2	Premature fast Gc banking bypasses validation
Profile 3	Rarely reaches this event
Profile 5	Validation works, but output does not feed into slow Gc

16. Transition Event 5: Bifurcation Threshold Crossing

The new structure crosses the bifurcation threshold.

Mutual-information dominance resolves the bifurcating dynamic for this specific structure, compiling it into fast Gc as an SR automation with documented boundary conditions.

Simultaneously, the structure enters the slow Gc gist extraction pathway.

The next loop iteration begins from a representationally richer state.

This is where the spiral advances.

Profile 5's failure occurs here: fast Gc compilation completes, but slow Gc installation fails before the next loop iteration begins.

Part IV — Complete Loop Phases

17. Phase 1: GP Regime Entry

All profiles enter this phase, but starting quality differs.

Profile	Entry condition
Profile 1	Rich slow Gc schematic geometry reshapes search
Profile 4	Thin SR substrate limits available depth
Profile 5	Prior iterations have not enriched the search space

The Layer 4 profile determines starting conditions for all subsequent phases.

18. Phase 2: SR Map Construction and Horizontal Tuning

The entropy arm finds a candidate relational structure.

The mutual-information arm validates task relevance.

The SR map is built and tuned horizontally across surface variations.

This is the power-law learning segment.

Profile	Phase 2 pattern
Profile 2	Steep but shallow segment
Profile 3	Shallow and noisy segment
Profile 1	Deep and efficient segment
Profile 4	Efficient but structurally limited segment

19. Phase 3: Asymptote and Bifurcation Response

This phase includes Transition Events 1 and 2.

It is the most diagnostically informative phase for Layer 4 assessment.

Profile-specific α stability signatures are most clearly expressed here.

20. Phase 4: Higher-Order Structure Emergence and Validation

This phase includes Transition Events 3 and 4.

The portability validation cascade discriminates between:

- genuine higher-order structure
 - local solutions
 - premature banking
 - non-convergent exploration
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21. Phase 5: Dual Gc Banking and Spiral Advance

This phase includes Transition Event 5.

Both Gc pathways activate:

- fast Gc compilation
- slow Gc gist extraction

The spiral property is the long-term intelligence-building function of the loop.

It is the primary differentiator between training that builds genuinely transferable intelligence and training that merely improves task performance.

22. Phase 6: Loop Integration and Next Iteration Preparation

The new fast Gc automation becomes available as a building block.

The slow Gc schema update reshapes the search space.

The GP regulation subsystem recalibrates its α stability baseline.

The Layer 4 profile may partially update.

Sustained profile-targeted training can produce genuine profile shifts, measurable in:

- transition event timing
- depth of structures emerging at Transition Event 3
- portability of fast Gc
- schematic enrichment across loops

This profile plasticity is the neural correlate of genuine intelligence development rather than task-specific skill acquisition.

Part V — Training Diagnostic Summary by Profile

Profile	Primary constraint	Key training target	Stack entry point	Diagnostic marker
1: Balanced GP	Relational territory richness	Novel cross-domain complexity	Top of stack	Rapid cross-domain transfer without prompting
2: MI dominant	Premature entropy closure	Entropy injection at breakpoints	Middle stack with entropy forcing	Early asymptote with poor portability

3: Entropy dominant	Insufficient MI constraint	Consolidation and evaluation discipline	Base stack with control training	Rich exploration without consolidation
4: SR depth deficient	Shallow relational maps	Deep non-verbal relational n-back	Lower-middle stack	Rapid asymptote across all domains
5: Consolidation deficient	Slow Gc pathway failure	Cross-domain diversity and sleep	Middle stack with rest discipline	Portable strategies without schematic integration

Part VI — Integration with Trident-G

23. What Trident-G Adds to the GP / Zhang–Tang / SR Architecture

The Trident-G framework adds the allostatic grounding of GP dynamics.

The GP architecture describes the bifurcating dynamic in representational and informational terms:

- entropy arm
- mutual-information arm
- SR temporal horizons
- FPCN-A/B balance
- α stability

Trident-G adds the organism-level energetic and regulatory substrate within which this dynamic operates.

The key additional concept is the F★ defended band.

The GP regime corresponds to the Ψ -band: the narrower operational corridor within the defended band where near-critical dynamics are realised and where fluid reasoning, controlled action, and creative reconfiguration become possible.

24. Why the Allostatic Grounding Matters

The earlier GP architecture implicitly assumed that the system is always capable of engaging the GP bifurcating dynamic.

Trident-G makes explicit that this capacity is itself regulated.

The ability to sustain FPCN-A/B competition varies with:

- stress
- health
- sleep
- developmental period
- meta-efficacy
- coherence
- niche coupling
- allostatic state

The Ψ -band's width and accessibility are shaped by the meta-parameters:

- η — meta-efficacy
 - χ — coherence
 - λ — niche coupling
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Part VII — Mapping Between Frameworks

25. Core Variable Mapping

Trident-G quantity	GP / Zhang–Tang / SR interpretation
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$\mathbf{x} = \mathbf{F} - \mathbf{F}^\star$	α stability monitor output; deviation from the GP-ready corridor
$\xi = \mathbf{G} - \mathbf{F}^\star$	FPCN-A mutual-information constraint level; active bracing or endorsed challenge
$\epsilon = \mathbf{F} - \mathbf{G}$	allostatic prediction error; realised demand versus expected burden
δ	explore—exploit tilt; FPCN-A/B balance
η	capacity to sustain bifurcating dynamics under challenge
χ	integration quality of FPCN-A/B coupling
λ	niche coupling; quality of link between internal GP dynamics and environmental structure

26. Mapping of Mismatch

$\epsilon > 0$: **Under-braced**

Reality is harsher than expected.

This drives entropy-arm reactivation and corresponds to Transition Event 2: the GP bifurcation response.

The system is forced into exploratory reconfiguration because current SR automations are insufficient.

$\epsilon < 0$: **Over-braced**

Reality is easier than expected.

This permits fast Gc compilation and corresponds to Transition Event 5: bifurcation threshold crossing.

Mutual-information dominance resolves the dynamic into stable automation.

27. Dopaminergic Implementation of Explore–Exploit Tilt

Trident-G's δ maps onto FPCN-A/B balance.

Dopaminergic channel	Trident-G function	G-loop function
D1-dominant Go	Better-than-expected outcomes consolidate current policy	Mutual-information constraint and fast Gc compilation
D2-dominant NoGo	Worse-than-expected outcomes reopen exploration	Entropy arm reactivation and relational search

The dopaminergic mechanism provides a biological implementation of the Zhang–Tang entropy / mutual-information balance.

Part VIII — Trident-G Cusp Geometry

28. Cusp Geometry as Bifurcation Topology

Trident-G's cusp catastrophe geometry gives a precise description of the bifurcation topology that the G loop's qualitative account requires.

The cusp potential is:

$$V(x; \xi, \delta) = x^4 - \xi x^2 - \delta x$$

This gives the bifurcating dynamic a precise geometric form with three identifiable regime regions corresponding to the cusp surface's structure.

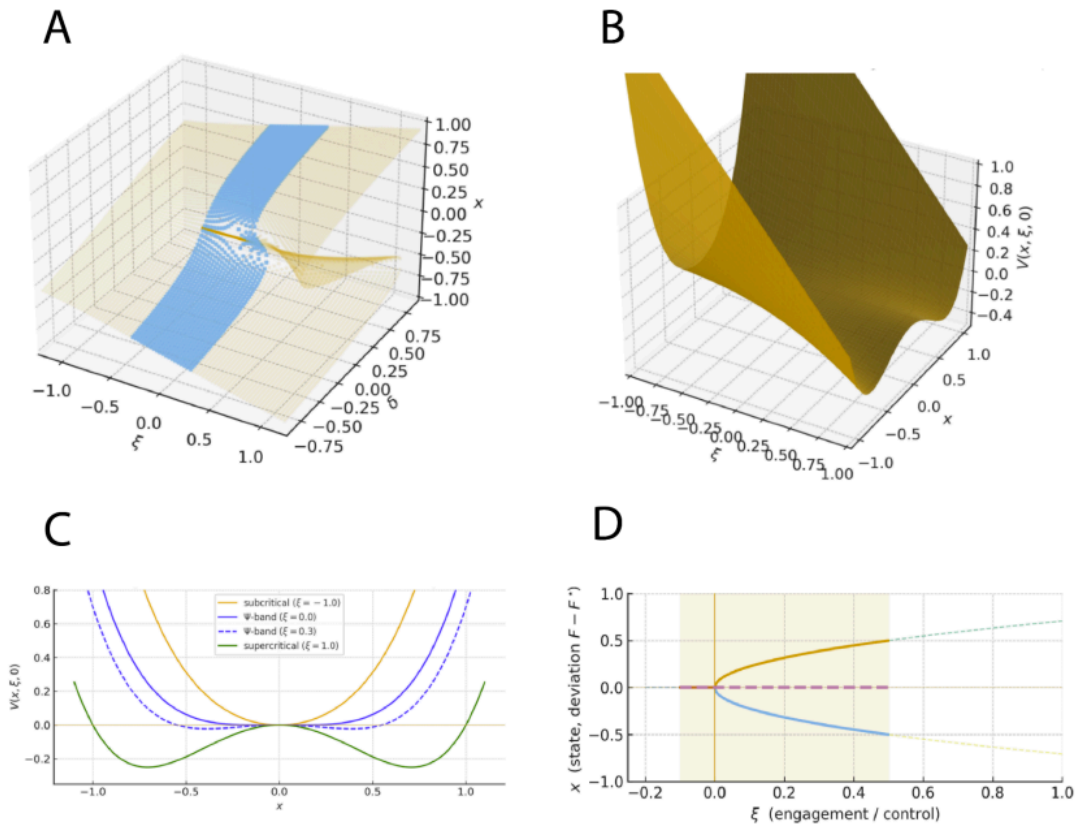


Figure 1. Trident-G cusp geometry and Ψ -band operating corridor. (A) Equilibrium cusp surface in (ξ, δ, x) , derived from $V(x; \xi, \delta) = x^4 - \xi x^2 - \delta x$, where $(x = F - F^*)$ indexes state deviation from the defended corridor, $(\xi = G - F^*)$ indexes engagement/control, and (δ) indexes explore–exploit tilt. The blue band schematically marks the near-critical Ψ operating corridor around the central fold region, where transitions between low-cost G_c /Autopilot execution and high-engagement G_f reconfiguration remain flexible and reversible. (B) Normalised cusp potential $V(x; \xi, 0)$ at zero tilt, showing the transition from a single-well to a double-well landscape as (ξ) increases. (C) One-dimensional cross-sections of the potential for different engagement levels (ξ) , illustrating the subcritical single-well regime, the near-critical Ψ -band regime, and the supercritical double-well regime. (D) Equilibrium branches $(x^*(\xi))$ in the (ξ, x) plane at $(\delta = 0)$, highlighting the Autopilot/ G_c shaft for $(\xi < 0)$, the shaded near-critical Ψ -band around $(\xi \approx 0)$, and the emergence of two high-engagement branches that can be biased by (δ) toward exploration- or exploitation-dominated G_f modes.

29. The Three Cusp Regimes and Their G-Loop Interpretations

29.1 Subcritical Single-Well Regime ($\xi < 0$)

When engagement ξ is negative — endorsed challenge is below the defended band — the cusp potential has a single shallow well near $x \approx 0$.

G-loop interpretation: This is the compiled G_c / SR automation regime. The bifurcating dynamic has already resolved. Mutual information has won, entropy has been suppressed, and the update distribution has thinned. The system operates efficiently within a single stable attractor — the Autopilot basin.

This corresponds to post-bifurcation subcritical G_c : model-free, subcortical-dominant, no longer internally bifurcating. It is not a low-engagement version of G_f but a qualitatively different dynamical mode.

29.2 Near-Critical Ψ -Band Regime ($\xi \approx 0$)

As ξ approaches zero from either direction, the central basin flattens. The cusp potential is shallow with low barrier heights. Switching between modes is easy and metabolically inexpensive.

G-loop interpretation: This is the GP / G_f regime — the sustained bifurcating dynamic. The system is neither fully committed to exploitation nor to exploration. The FPCN-A/B competition is genuinely alive. The α stability monitor shows healthy heavy-tailed power-law statistics. The entropy arm and mutual-information arm are in genuine dynamic competition.

This is the region of maximum flexibility and minimum commitment: the operational Ψ -band where G-loop iterations generate genuine relational learning rather than mere performance optimisation within an existing schema.

29.3 Supercritical Double-Well Regime ($\xi \gg 0$)

When engagement ξ is strongly positive, the central point becomes unstable and two deep outer wells appear at larger values of $|x|$. The barrier between wells is high, transitions between them are metabolically costly, and the system is committed to one of two high-engagement modes.

G-loop interpretation: This is the supercritical fragmentation failure mode — the entropy arm operating without sufficient mutual-information constraint. The system is generating broad search without convergence. The two wells correspond to two qualitatively distinct high-load Gf branches that can be biased toward exploration-dominated or exploitation-dominated processing when a small tilt $\delta \neq 0$ is present.

A small tilt δ specifying which of the two wells is occupied maps directly onto the FPCN-A/B balance within the GP regime — whether the system is tilted toward the mutual-information arm's exploitation-dominated processing or the entropy arm's exploration-dominated processing at any given moment.

30. Meta-Parameter Modulation of Cusp Geometry

The cusp potential includes meta-parameters that reshape the landscape:

$$V(x; \xi\star, \xi, \eta, \chi, \lambda) = b(\xi\star, \eta)x^4 - a(\xi\star, \chi)\xi_{\text{eff}}(t)x^2 - \delta(t)x$$

Where:

- $\xi_{\text{eff}}(t) = \lambda(t)[\xi\star(t) + \tilde{\xi}(t)]$ — effective engagement after niche coupling
- $a(\xi\star, \chi)$ — quadratic coefficient: controls how sharply engagement splits Autopilot versus Gf regimes
- $b(\xi\star, \eta)$ — quartic coefficient: controls well depth and stiffness

How each meta-parameter shapes the landscape

η (meta-efficacy): Modulates the quartic coefficient b . Higher η effectively flattens the potential — shallower wells, more freedom to move between regimes, lower cost of operating near the Ψ -band fold. Lower η makes wells deeper and stiffer — the system feels less safe operating near the bifurcation point and tends to collapse into either the Autopilot well or a high-engagement well and stay there. This explains why damaged meta-efficacy (chronic failure experiences, PTSD, depression) makes the Ψ -band harder to access even when allostatic load is technically within range.

χ (coherence): Modulates the quadratic coefficient a . High χ gives steeper, more clearly differentiated wells — the Autopilot and Gf modes are more distinct and switching between them is more decisive. Low χ flattens the splitting — the system cannot clearly differentiate between modes and transitions become blurred and unreliable. This is the GP architecture's coherence requirement: without sufficient

integration quality in the FPCN-A/B coupling, the bifurcating dynamic cannot maintain stable competition between the two arms.

λ (niche coupling): Amplifies ξ_{eff} — how much of the endorsed challenge level actually reaches the system from the niche. High λ means environmental demands cut deeper grooves in the cusp landscape for a given $G - F^{\star}$. Low λ means the system's internal GP dynamics are partially decoupled from environmental structure — the entropy arm is searching a space that doesn't accurately reflect the niche's relational demands, reducing the portability of whatever structures emerge from the bifurcation.

Training, trauma, and meta-parameter shifts

Training, illness, and environment can move and stretch the Ψ -band by shifting these meta-parameters:

- Successful training and robust niches widen the Ψ -band (increase η and λ , stabilise χ)
- Chronic stress and impoverished niches narrow it (reduce η , destabilise χ , reduce λ)
- These shifts appear as concrete changes in the cusp landscape geometry — how much of the defended band is reachable and stable as a high-functioning near-critical region

31. Tonic and Phasic Engagement: The $\xi^{\star}$ / $\tilde{\xi}$ Distinction

A critical distinction for individual differences and training design is the decomposition of engagement ξ into two components:

Tonic engagement bias ($\xi^{\star}$):

$$\xi^{\star}(t) = G^{\star}(t) - F^{\star}(t)$$

The long-run challenge stance in a given niche. This is a trait-like variable — how boldly the system habitually operates near the Ψ -band fold. High $\xi^{\star}$ means the system is chronically set toward challenge-seeking and high engagement. Low $\xi^{\star}$ means the system chronically expects and endorses low demand.

Phasic engagement update ($\tilde{\xi}$):

$$\xi(t) = G(t) - G^\star(t)$$

The short-term adjustment around the tonic stance. For this specific episode, I will push harder, or for now, I am backing off. This is a state-like variable that can shift on timescales of seconds to hours.

Full engagement:

$$\xi(t) = \xi^\star(t) + \tilde{\xi}(t)$$

Why this distinction matters for the G loop

The five Layer 4 profiles differ primarily in $\xi^\star$ — their trait-level tonic engagement bias:

- Profile 2 (mutual information dominant) corresponds to chronically high $\xi^\star$ — the system is set toward exploitation and premature constraint application even before specific task demands are known
- Profile 3 (entropy dominant) corresponds to chronically low $\xi^\star$ — the system is set toward exploration but cannot apply sufficient constraint when it matters
- Profile 1 (balanced GP) corresponds to $\xi^\star$ calibrated near the Ψ -band optimum for the relevant domain

Training operates primarily through repeated phasic $\tilde{\xi}$ shifts — creating episodes that push the system beyond its current $\xi^\star$ baseline and demonstrating successful navigation. When these phasic successes accumulate, they update $G^\star$ (the baseline expected load) through prediction-error learning, gradually reshaping $\xi^\star$ itself. This is the mechanism of genuine profile shifts: sustained training that targets profile-specific constraints produces tonic changes in the challenge stance, not just state-level performance improvements.

The key training calibration principle follows directly: interventions should produce systematic competence-confirming negative mismatch ($\epsilon < 0$) — challenges that initially feel demanding, but turn out to be more manageable than expected. In Trident-G terms, this means realised burden is slightly lower than expected burden: $\epsilon = F - G < 0$.

This slightly negative mismatch teaches the system that it can meet challenge without leaving the Ψ -band. Repeated experiences of this kind should increase meta-efficacy η , reduce unnecessary bracing, and gradually widen the accessible Ψ -band.

Chronically negative ε , when produced by well-calibrated challenge rather than avoidance, confirms competence and expands the operational corridor. Chronically positive ε , where reality is repeatedly harsher than expected, produces overload, erodes meta-efficacy, and narrows the Ψ -band.

31.1 Defining ξ : Objective and Subjective Characterisations

Because ξ determines whether the system is operating in, above, or below the Ψ -band — and because its tonic component $\xi^\star$ is the primary differentiator of the five individual differences profiles — it warrants explicit definition at both the objective and subjective levels. These two characterisations are complementary: the objective definition specifies what ξ indexes in terms of measurable system states; the subjective definition specifies how ξ is experienced from the inside, which is essential for clinical application and training design.

Objective definition

ξ is the signed distance between the system's current endorsed challenge level and its allostatic set-point — the degree to which the system is mobilised above or below its sustainable near-critical operating baseline. It is a preparatory mobilisation index, not a response to actual load. This distinction is critical: ξ indexes the stance the system has adopted toward anticipated demand, not the demand that has actually arrived (which is F) and not the gap between them (which is ε).

$\xi > 0$: The system is mobilised above its set-point. More physiological, attentional, and energetic resource is committed than baseline sustainable operation requires. Measurable correlates include elevated tonic norepinephrine and dopamine tone, increased prefrontal-subcortical coupling, elevated heart rate variability baseline, heightened corticospinal excitability, and increased default-mode suppression at rest. These are not responses to actual load — they are the preparatory stance the system has adopted toward anticipated load.

$\xi = 0$: Mobilisation is at the set-point. The system is in its sustainable near-critical corridor, neither over-prepared nor under-prepared for its typical environmental demands. This is the objective correlate of the Ψ -band's lower boundary.

$\xi < 0$: The system is mobilised below its set-point. Less resource is committed than the defended baseline requires. Measurable correlates include reduced tonic arousal, decreased prefrontal engagement, increased default-mode dominance at rest, reduced corticospinal excitability, and low-variability SR automation dominance. This is the allostatic signature of the dark-room or compiled-Gc regime.

Subjective definition

Subjectively, ξ is the felt sense of how much this situation is asking of me relative to what I consider normal and manageable — specifically the endorsement of that demand level, not merely its perception. This distinction between perception and endorsement is diagnostically important. A person can perceive high demand (high F) without endorsing it, producing a dissociated or avoidant response. Conversely, a person can endorse high challenge (high ξ) in an objectively low-demand environment — the hypomanic state, the competitive athlete warming up, the person who finds rest threatening. ξ is the endorsed challenge level, not the perceived one.

$\xi \gg 0$ feels like: chronic readiness that won't switch off; the sense that something important is at stake or that threat is imminent even when nothing is happening; difficulty relaxing; the background hum of being mobilised without a target; anticipatory tension that feels necessary rather than optional. In GAD this is experienced as pervasive worry. In high-functioning high-achievers it is experienced as drive or ambition — the chronic endorsement of high challenge as the normal operating mode. The key subjective signature is that the mobilisation feels justified or necessary, not excessive: the system endorses it.

$\xi \approx 0$ feels like: appropriate engagement — neither restless nor flat; the sense that the current situation is demanding enough to be interesting without being threatening; the capacity to be fully present without bracing. The subjective signature is the absence of the endorsement gap in either direction — neither straining toward challenge nor defending against it. This is the felt quality of what people describe as flow conditions at their lower boundary.

$\xi \ll 0$ feels like: flatness, disengagement, the sense that nothing is worth the effort; a world that makes no demands and offers no purchase; the specific quality of anhedonia where potential rewards register intellectually but produce no preparatory mobilisation. The subjective signature is the absence of endorsed challenge — not necessarily the absence of perceived demand, but the failure of demand to recruit mobilisation.

The tonic/phasic decomposition subjectively

$\xi \star$ (tonic) is the subjective default stance toward the world — the background level of challenge the person considers normal and implicitly endorses. This is what people mean when they describe someone as driven, anxious, laid-back, or avoidant as a personality characteristic. It is experienced not as a choice but as the texture of ordinary life: how demanding things normally feel, how much effort feels appropriate to expend by default. $\xi \star$ is not accessible to direct introspection as a parameter — it is the background against which all experience is measured, invisible precisely because it defines the zero point.

ξ (phasic) is the subjective decision to lean in or pull back from the tonic stance — the felt sense of “I am going to push harder on this” or “I need to back off right now.” This is experienced as volitional, or at least accessible to deliberate influence, in a way that $\xi\star$ typically is not. Motivational interviewing, challenge framing, and pre-performance routines all operate primarily on ξ . The phasic component is the lever that training interventions can move on short timescales; the tonic component is what shifts slowly through accumulated evidence that the world is more or less demanding than the current $G\star$ baseline assumes.

The critical clinical implication of the subjective definition

Patients do not experience ξ as a deviation from a set-point — they experience it as the set-point itself. A person with chronic high $\xi\star$ (GAD, PTSD hypervigilance, perfectionism) does not feel over-mobilised relative to their normal — their normal is over-mobilised. The chronic endorsement of high challenge feels like appropriate attentiveness to a genuinely demanding world. The subjective signature of pathological $\xi\star$ elevation is therefore not “I feel too anxious” but “the world is genuinely this demanding and I am appropriately prepared for it.”

This is why $\xi\star$ revision is so therapeutically difficult — it requires the system to update its endorsed challenge baseline, which means accepting that the world is less demanding than it has been treating it as. The existing $G\star$ architecture experiences this as a dangerous reduction in preparedness rather than as appropriate recalibration. The subjective correlate of successful $\xi\star$ downward revision is therefore not relief but initially vulnerability — the felt sense of being underprotected, before the new baseline consolidates as normal. Effective interventions must anticipate and work with this vulnerability window rather than treating it as treatment failure.

Conversely, a person with chronically low $\xi\star$ (depression, chronic disengagement) does not feel under-mobilised — they feel that the world genuinely offers nothing worth mobilising for. The pathology is invisible from the inside because the endorsed challenge level matches the perceived opportunity level: both are low, so no gap is felt. This is why depression is so resistant to motivational intervention: an appeal to ξ is ineffective when $\xi\star$ has established a baseline where nothing warrants endorsement. Effective intervention must address the schematic G_c geometry that is constraining the entropy arm’s search space — the active representational substrate of the low $\xi\star$ — not merely the motivational stance itself.

Part IX — Operational Regimes: A Unified Two-Axis Classification

32. Two Classification Systems and Why Both Are Needed

Two classification systems apply to the operational regimes of the G loop. They cut the space differently and must be distinguished clearly.

The Trident-G operational regimes (well-regulated, over-braced, under-braced, dark-room/adventure) describe the allostatic pathways into different states — how the system's (x , ξ , ϵ) configuration is being driven. They describe the forces acting on the system.

The dynamical state classification (subcritical flat, in the zone, supercritical spun-out, supercritical locked-in) describes where the system is on the cusp surface — what the system's internal dynamics look like once it is in a given region. They describe the state itself.

Both are needed because the same dynamical state can be reached by different allostatic routes, and the same allostatic route can produce different dynamical states depending on the system's pre-existing ξ ★ bias, cusp geometry, and meta-parameter configuration.

33. The Two-Axis Dynamical State Classification

The dynamical state classification cuts the cusp surface along two independent axes:

Axis 1: ξ — subcritical versus supercritical Whether the system is below the bifurcation threshold (single well, no GP dynamics) or above it (double well, GP dynamics available but potentially unstable).

Axis 2: δ — which well of the supercritical double-well When supercritical, whether the entropy arm or the mutual-information arm is dominating — which branch the system is occupying.

This yields four distinct dynamical states:

33.1 Subcritical Flat

Cusp position: $\xi < 0$. Single shallow Autopilot well. δ is irrelevant — there is only one well to occupy.

Dynamical signature: No GP bifurcating dynamic. The bifurcation has either not been entered or has fully resolved. Update distribution thin. α stability below the GP range. The system is operating in pure SR automation mode — model-free, subcortical-dominant, efficient within its compiled strategy but unable to generate genuine relational reconfiguration.

G-loop state: Loop cannot initiate productively. No prediction error large enough to trigger Transition Event 2. The system cycles through existing fast Gc automations without spiral advance.

Relation to profiles: The chronic failure state of Profile 5 (consolidation deficient — loop cycles without spiralling). Also the intended destination for compiled Gc structures — subcritical flat is the correct regime for automated expertise, pathological only when the system cannot exit it.

Allostatic routes in: Dark-room proper ($\xi \ll 0$, both actual and expected load low, ε small); chronic depression (F★ has drifted, system is below Ψ -band); or successful Gc compilation (the system has correctly resolved the bifurcation for a specific structure and is executing it efficiently).

33.2 In the Zone

Cusp position: $\xi \approx 0$. Near-critical Ψ -band region. Central basin shallow with low barrier heights. δ small and balanced.

Dynamical signature: Full GP bifurcating dynamic active. FPCN-A/B competition genuine and productive. α stability maintained in the healthy heavy-tailed range. The I - β product is approximately stable — the Zhang-Tang invariant holds. Neither entropy arm nor mutual-information arm dominates long enough to collapse the dynamic. Switching between modes is easy and metabolically inexpensive.

G-loop state: All five phases accessible. Loop can initiate, sustain SR tuning, complete the bifurcation response, generate higher-order structures, and complete the dual Gc banking sequence. This is the productive iteration state.

Relation to profiles: The operating state of Profile 1 (balanced GP). The training target for all other profiles.

Allostatic routes in: Well-regulated Ψ -band regime (x and ξ both near zero, ε small); adventure sub-regime ($x \ll 0$ but $\xi \geq 0$, strongly negative ε driving challenge-seeking); calibrated training challenge (ε consistently slightly negative, building η and widening the Ψ -band).

33.3 Supercritical Spun-Out

Cusp position: $\xi \gg 0$, δ tilted toward the entropy-dominant well (exploration-dominated Gf branch).

Dynamical signature: Entropy arm dominating without sufficient mutual-information constraint. FPCN-B broadly activated, FPCN-A engagement intermittent or weak. Update distribution develops extreme heavy tails without convergence. α exceeds the optimal range — supercritical dynamics rather than GP regime maintenance. β remains high while λ fails to rise. Rich generative search is occurring but cannot resolve into validated relational structure.

G-loop state: Transition Event 3 (higher-order structure emergence) is delayed or fails. Many candidate relational configurations are generated but cannot be selected among. Fast Gc banking is rare. The bifurcation threshold is rarely crossed cleanly. Slow Gc gist extraction is rich in diversity but produces schematic geometry that is broad and associatively rich but poorly calibrated to task-relevant structure — creative but unreliable.

Relation to profiles: The characteristic operating state of Profile 3 (entropy dominant). Also the acute failure mode under genuine under-bracing when D2-mediated exploration opens without the mutual-information constraint to direct it productively.

Allostatic routes in: Two distinct pathways reach this state:

- *Acute under-bracing* ($\varepsilon > 0$ at high magnitude — reality far harsher than expected, emergency exploration activated without productive constraint)
- *Chronic low ξ ★ trait bias* (Profile 3's stable tonic engagement bias — the system is chronically set toward exploration regardless of acute allostatic state)

These are mechanistically different. Acute spun-out resolves when the allostatic mismatch resolves. Chronic spun-out (Profile 3) persists because it reflects a stable

FPCN-A/B balance bias — it requires the MFT-M mutual-information constraint training rather than simply stress reduction.

33.4 Supercritical Locked-In

Cusp position: $\xi \gg 0$, δ tilted toward the mutual-information dominant well (exploitation-dominated Gf branch).

Dynamical signature: Mutual-information arm dominating without sufficient entropy arm activation. FPCN-A strongly activated, FPCN-B search prematurely closed. Update distribution flattens toward thin tails as the system over-commits to its current constraint schema. α falls below the optimal heavy-tailed range — the system is operating subcritically during what should be the GP bifurcation response phase. λ rises quickly while β collapses.

G-loop state: Transition Event 2 (GP bifurcation response) is truncated. The entropy arm activates briefly but FPCN-A recaptures the dynamic before genuine higher-order structure can emerge. Fast Gc banking occurs prematurely — structures are compiled before portability validation cascade is complete. Wrapper-specific automations accumulate. The loop appears productive but generates brittle, poorly portable Gc.

Relation to profiles: The characteristic operating state of Profile 2 (mutual information dominant). Also the failure mode of OCD (extreme locked-in), PTSD's hypervigilant exploitation well, and rigid expertise that cannot reconfigure under novel conditions.

Allostatic routes in: Two distinct pathways reach this state:

- *Acute over-bracing* ($\xi \gg 0$, $\varepsilon < 0$, the system is braced for threat and applying tight constraint — high $G\star$ relative to actual F, chronic hypervigilance)
- *Chronic high $\xi\star$ trait bias* (Profile 2's stable tonic engagement bias — the system is chronically set toward exploitation and constraint regardless of acute allostatic state)

Again, these are mechanistically different. Acute locked-in resolves when the over-bracing resolves and $G\star$ is revised downward. Chronic locked-in (Profile 2) persists because it reflects a stable FPCN-A/B balance bias — it requires specific entropy injection training at the bifurcation response phase rather than simply reducing arousal.

Important non-equivalence: A system with low χ (poor FPCN-A/B coupling coherence) may present as spun-out even under conditions of over-bracing, because the mutual-information arm cannot maintain coherent constraint even when

over-activated. Conversely, a system with very deep existing cusp wells (high accumulated Gc rigidity) may present as locked-in even under under-bracing, because it defaults to its most familiar exploitation strategy rather than generating novel exploration. The dynamical state and the allostatic pathway are therefore partially independent and both must be assessed for accurate diagnostic and training prescription.

34. Unified Mapping: Dynamical States × Allostatic Pathways

The two classification systems interact as follows:

Dynamical state	Primary allostatic route	Secondary allostatic route	Profile correspondence
Subcritical flat	Dark-room ($\xi \ll 0, \varepsilon \approx 0$)	Depression (F★ drift)	Profile 5 failure; compiled Gc (intended)
In the zone	Well-regulated ($x \approx 0, \xi \approx 0$)	Adventure ($x \ll 0, \xi \geq 0$)	Profile 1
Supercritical spun-out	Chronic low ξ ★ trait bias	Acute under-bracing ($\varepsilon \gg 0$)	Profile 3
Supercritical locked-in	Chronic high ξ ★ trait bias	Acute over-bracing ($\xi \gg 0, \varepsilon < 0$)	Profile 2

35. The Trident-G Operational Regimes as Allostatic Pathways

The Trident-G operational regimes — originally formulated as the primary classification — are now understood as the allostatic pathway layer: they describe the forces driving the system toward or away from each dynamical state.

35.1 Well-Regulated Ψ -Band Regime

Conditions: $x \approx 0$, $\xi \approx 0$. $|x|$, $|\xi|$, $|\varepsilon|$ all small.

Allostatic function: Maintains the system in or near the in the zone dynamical state. The cusp is multi-stable but not excessively stiff. The system can engage Gf when needed and relax back to Autopilot without large allostatic swings.

Training relevance: The target allostatic configuration for all productive G-loop iteration. Training should be calibrated to maintain this regime — specifically in the adventure sub-region where $\xi \geq 0$ while F remains manageable.

35.2 Over-Braced Regime

Conditions: $\xi \gg 0$, x smaller. $\varepsilon = x - \xi < 0$ (reality easier than expected).

Allostatic function: Primary pathway into the supercritical locked-in dynamical state. The mutual-information arm is chronically over-activated relative to actual prediction-error signal. Compresses entropy arm reach. Deep, steep wells. High mobilisation cost.

Over time, repeated negative ε should drive $G\star$ downward. If this revision fails — as in anxiety disorders and PTSD — the system remains chronically over-braced, Ψ -band narrows through sustained metabolic cost, and η erodes.

Training relevance: The training prescription of entropy injection is specifically designed to create repeated negative ε in the bifurcation response phase, demonstrating the system can tolerate exploration without catastrophic outcomes — gradually revising $\xi\star$ downward and widening the Ψ -band.

35.3 Under-Braced Regime

Conditions: $x \gg 0$, ξ small. $\varepsilon = x - \xi > 0$ (reality harsher than expected).

Allostatic function: Acute pathway into supercritical spun-out (emergency exploration) or paradoxically into locked-in (emergency exploitation in systems with

deep existing wells). The positive prediction error signal that should drive productive Transition Event 2 is present but at excessive magnitude — the system enters emergency mode rather than productive bifurcation response.

Chronic under-bracing progressively erodes η through sustained overload, narrows the Ψ -band, and degrades mutual-information arm precision.

Training relevance: The regime the training calibration principle specifically avoids. The appropriate training stress is achievable challenge — tasks consistently in the range where $\varepsilon < 0$ is the typical outcome.

35.4 Dark-Room and Adventure Regimes

Dark-room conditions: $F \ll F_\star$, $\xi \ll 0$. Both actual and expected load are low. $\varepsilon \approx 0$ — nothing pushes the system out.

Allostatic function: Pathway into and maintenance of the subcritical flat dynamical state. The cusp flattens around a single low-engagement Autopilot well. No G-loop initiation possible. Profile 5's chronic failure in its sustained form.

Adventure conditions: $F \ll F_\star$, $\xi \geq 0$. Actual load low but endorsed challenge high. $\varepsilon = x - \xi \ll 0$ — strongly negative, driving challenge-seeking.

Allostatic function: Productive sub-regime that maintains near-critical dynamics without full bifurcation activation. The system is in the lower Ψ -band — optimal for slow G_c gist extraction. The adventure regime is what good consolidation intervals look like allostatic-ally: near-critical enough for rich statistical structure, stable enough for consolidation to complete.

Training relevance: Genuine consolidation intervals should target the adventure sub-regime — not pure dark-room inactivity but deliberate low-load engagement maintaining near-criticality without new prediction-error override. Sleep achieves this through oscillatory dynamics that keep the system near-critical while the slow G_c pathway installs gist as schematic geometry.

Part X — Separation of Timescales

36. Three Layers of Change

A critical architectural distinction for training design is the separation of the framework's variables into three distinct timescale layers.

36.1 Fast and Semi-Fast: Meta-Parameters and Expectations

Variables: η , χ , λ , G , $\xi = G - F\star$, δ

Timescales:

- seconds to minutes: neuromodulators, attention shifts, momentary belief changes
- hours to days: standard synaptic plasticity, therapeutic conversations, feedback
- weeks to months: training, psychotherapy, lifestyle implementation

These meta-parameters effectively reconfigure how the system rides the cusp and occupies the Ψ -band without needing to move the underlying metabolic criticality threshold. They change:

- how strongly engagement carves distinct Autopilot versus Gf wells
- how often the system leans into exploration versus exploitation
- how safe it feels operating near the cusp folds

Placebo and belief effects operate primarily at this layer — changing η , G , and the interpretation of surprises rather than the hardware itself.

G-loop implication: Most acute training effects — session-to-session performance improvements, therapeutic gains, challenge calibration adjustments — operate at this layer. They reshape how the system uses its existing critical band and Gc substrate. They do not directly expand the substrate itself.

36.2 Intermediate: Self-Organising Gc Structure

Variables: The structure of the generative model — semantic, relational, and skill patterns built through repeated operation near criticality

Timescale: Weeks to months through the slow Gc gist extraction and cortical consolidation pathway

This is the active Gc layer: schematic geometry that is neither a fast meta-parameter shift nor a fixed physiological property, but an intermediate-rate self-organising structure emerging from repeated G-loop iteration.

The slow Gc pathway installs relational invariants as cortical representational geometry. Once installed, these invariants actively reshape the search space available to the entropy arm in subsequent loop iterations — this is what distinguishes the self-enriching spiral from flat cycling.

G-loop implication: The spiral advance of the G loop operates at this timescale. Individual loop iterations produce fast Gc compilations (sessions to weeks). These feed the slow Gc gist extractor (weeks to months). The schematic geometry that results actively enriches the starting conditions for future iterations. Training programs shorter than this timescale cannot produce genuine spiral advance — they can improve performance within the current schematic geometry without expanding it.

36.3 Slow: Physiology of Network Criticality

Variables: Anatomical connectivity, white-matter structure, vascular and metabolic capacity, baseline excitation–inhibition balance, endocrine milieu, long-term health

Timescale: Months to years — development, ageing, chronic disease, intensive long-term training

These determine:

- the location of the near-critical window in terms of energy or load (the feasible range for $F^\star$)
- the intrinsic breadth of the underlying Griffiths phase

G-loop implication: This layer sets the ultimate limits on where the Ψ -band can sit and how wide it can be. It defines the long-run boundaries within which both meta-parameter adjustment and self-organising Gc construction must operate. Development, lifestyle, ageing, and chronic disease shift these boundaries slowly but significantly. Very long-term training programs — years of sustained practice — can genuinely expand this substrate, but single training interventions cannot.

36.4 Why This Separation Matters for Training Design

Timescale	What changes	What does not change
Fast (minutes–days)	$\eta, \chi, \lambda, G, \xi, \delta$	Gc schematic geometry, criticality substrate
Intermediate (weeks–months)	Slow Gc schematic geometry	Criticality substrate
Slow (months–years)	Physiological criticality thresholds	—

A training program that produces rapid performance gains but no change in the slow Gc schematic geometry (no spiral advance) is operating exclusively at the fast layer — adjusting meta-parameters on a fixed substrate. It will not produce far transfer.

A training program that produces genuine spiral advance — measurable enrichment of the schematic geometry across loop iterations — is operating at the intermediate layer. This requires adequate loop iterations with sufficient episodic diversity and adequate consolidation intervals. It will produce genuine far transfer.

A training program that expands the physiological criticality substrate — genuinely widening the Ψ -band at the hardware level — requires years of sustained near-critical operation and healthy lifestyle support. It corresponds to genuine long-term intelligence development rather than skill acquisition.

Part XI — The Computational-Level Definition of General Intelligence

37. General Intelligence as Ψ -Band Navigation Capacity

At Marr's computational level, general intelligence g can be defined precisely within the Trident-G framework.

Let $\text{Occ}\Psi(t)$ indicate whether the current state $(x, \xi, \delta, \eta, \chi, \lambda)$ lies within the Ψ -band.

Let $\langle \text{Occ}\Psi \rangle$ be the time-average of Ψ -band occupancy under standardised challenges.

Let $\mathcal{G}$ denote the set of recurrent Ψ -band trajectories — G-loops — that the organism can reliably realise and re-enter.

Working definition of g :

g is an increasing function of (i) $\langle \text{Occ}\Psi \rangle$ and (ii) the volume or diversity of $\mathcal{G}$ in $(x, \xi, \delta, \eta, \chi, \lambda)$ -space.

Informally: more intelligent organisms spend more time in their Ψ -band and can access a richer repertoire of near-critical loops without falling out of their viability band.

37.1 What This Definition Captures

$\langle \text{Occ}\Psi \rangle$ — time in the Ψ -band: This captures the sustained maintenance of GP dynamics under real-world challenge. A system that frequently drops out of the Ψ -band under moderate load has lower g than one that maintains near-critical dynamics across varied and demanding conditions. This corresponds to the robustness dimension of intelligence — the ability to maintain GP operation despite perturbation.

Volume of $\mathcal{G}$ — diversity of accessible G-loops: This captures the richness of the relational territory the system can navigate while remaining in the Ψ -band. A system that can sustain G-loops across many domains, timescales, and levels of abstraction has higher g than one whose GP dynamics are confined to a narrow representational territory. This corresponds to the flexibility dimension of intelligence — the ability to generate genuinely novel relational structures across diverse domains.

The interaction: The definition requires both. High $\langle \text{Occ}\Psi \rangle$ with narrow $\mathcal{G}$ is robust but not flexible — the system stays near-critical but only in familiar territory. Wide $\mathcal{G}$ with low $\langle \text{Occ}\Psi \rangle$ is potentially flexible but unreliable — the system can generate novel structures but cannot sustain the dynamics needed to validate and compile them. Peak general intelligence requires sustained Ψ -band occupancy across a rich and diverse G-loop repertoire.

37.2 How the G-Loop Builds g

Each complete G-loop spiral advances along both dimensions of the g definition:

- Ψ -band occupancy increases because the slow Gc schematic geometry enriches the representational substrate, making future loop entry from a richer starting state and requiring less recovery time after perturbation
- G-loop repertoire expands because the fast Gc compilations add new SR automations as structured building blocks, and the slow Gc gist extraction installs cross-domain invariants that allow the entropy arm to reach genuinely novel relational territories on subsequent iterations

Training that produces genuine spiral advance — full loop iterations with dual Gc banking and adequate consolidation — therefore directly increases both components of g. Training that produces performance improvement without spiral advance increases neither.

37.3 Implications for the Individual Differences Profiles

The g definition specifies precisely what each profile lacks:

Profile	$\langle \text{Occ}\Psi \rangle$ limitation	$\mathcal{G}$ volume limitation
1: Balanced GP	Neither — limited only by relational territory available	Neither — limited only by novelty of challenges encountered
2: MI dominant	Ψ -band collapses during bifurcation response phase	G-loops confined to existing schema territory
3: Entropy dominant	Ψ -band destabilises into supercritical fragmentation	Nominally wide but G-loops rarely complete; repertoire is shallow
4: SR depth deficient	Ψ -band maintained but on thin substrate	G-loops complete correctly but reach only shallow invariants
5: Consolidation deficient	Ψ -band maintained within iterations but drops between them	G-loop repertoire does not expand across iterations; $\mathcal{G}$ is static

Part XII — Socially Extended Trident-G

38. The Group as a Trident-G Agent

The Trident-G framework applies not only to individual organisms but to groups, teams, and collectives that are sufficiently coupled to behave as semi-coherent agents.

When individuals are coupled through shared beliefs, norms, roles, joint policies, and interaction patterns, a group-level generative model emerges with its own quasi-Markov blanket relative to the wider environment.

The same core quantities can be defined at the group scale:

Quantity	Group interpretation
$F_{\text{group}}(t)$	Group-level allostatic load — overall stress on the collective, including resource scarcity, conflict, coordination cost, and risk exposure
$G_{\text{group}}(t)$	Expected group load — how demanding the group tacitly expects the shared niche to be
$F^{\star}_{\text{group}}(t)$	Defended group set-point — reference level of group strain compatible with sustainable near-critical operation
$G^{\star}_{\text{group}}(t)$	Baseline expected group load — slowly learned sense of how demanding life typically is for this group in this niche

From these:

$$x_{\text{group}}(t) = F_{\text{group}}(t) - F^{\star}_{\text{group}}(t)$$

$$\xi_{\text{group}}(t) = G_{\text{group}}(t) - F^{\star}_{\text{group}}(t)$$

$$\epsilon_{\text{group}}(t) = F_{\text{group}}(t) - G_{\text{group}}(t)$$

The group's Ψ -band is the region in $(F_{\text{group}}, G_{\text{group}})$ space where:

- collective load is neither chronically overwhelming nor chronically under-stimulating

- the group maintains high dynamic range in internal coordination
 - recovery from perturbations is high
 - multiple workable coordination modes are available
-

39. Group-Level Meta-Parameters

The group cusp geometry is shaped by group-level meta-parameters:

- η_{group} — collective meta-efficacy: the group's belief in its controllability and competence; built through demonstrated performance across varied challenges
- χ_{group} — group coherence: the integration quality of the group's shared generative model; built through trust, data sharing, and aligned objectives
- λ_{group} — niche coupling: the strength of connection between the group's internal models and its shared environment; built through genuine environmental engagement and feedback integration

These parameters reshape the group cusp geometry in the same way individual meta-states reshape the individual cusp — determining how deep and stiff the coordination wells are, how wide the group Ψ -band can be, and how readily the group can switch between exploitation and exploration modes.

40. Mechanisms of Group-Level Coupling

For a social Trident-G to be meaningful, individuals must not merely model each other but become dynamically coupled such that a shared higher-level structure emerges. Three coupling mechanisms support this:

Neural and physiological synchrony: Interpersonal synchrony in brain activity, heart rate, and autonomic signals during joint action, conversation, and cooperation. From a Trident-G perspective, this is what occurs when individuals' internal trajectories are pulled into generalised synchrony by reciprocal prediction and signalling.

Behavioural synchrony and joint rhythms: Walking in step, complementary speech rhythms, and aligned postures reduce the cost of mutual prediction. These effectively sharpen the group's cusp — producing clearer, more stable coordination modes — while lowering individual prediction errors.

Symbolic communication and shared narratives: Language, shared stories, rules, and norms allow the group to build and maintain a shared generative model that persists beyond any momentary interaction. This is the cultural substrate of G★group and F★group — the slowly evolving sense of what the group can handle and how demanding its niche typically is.

41. The GP Collective as Group-Level Trident-G

The GP collective business architecture we have described is a direct instantiation of socially extended Trident-G.

Each partner node is an individual Trident-G agent maintaining its own Ψ -band navigation, its own fast Gc SR automation store, and its own domain-specific G-loop dynamics.

The shared slow Gc gist store is the group's shared generative model substrate — the cultural and schematic geometry built from cross-node episodic diversity through the collective's gist extraction pathway. This corresponds to the symbolic communication and shared narratives mechanism at the cognitive infrastructure level.

The GP kernel's regulation function corresponds to the group's GP regulation subsystem — monitoring the collective's α stability, detecting when mutual-information dominance is suppressing node-level exploration, and triggering entropy injection through cross-domain challenge and novel partnership configurations.

The governance architecture corresponds to the group cusp geometry's meta-parameters:

- building η group through demonstrated collective performance
- building χ group through trust and data-sharing structures
- building λ group through genuine environmental engagement and feedback

The collective's Ψ -band is the organisational regime where:

- collective load is neither chronically overwhelming nor under-stimulating
- semi-independent node-level bifurcation dynamics are preserved
- recovery from environmental perturbations is high
- the collective can switch between exploitation-dominant coordination (scaling proven strategies) and exploration-dominant coordination (generating novel cross-domain configurations)

The socially extended Trident-G formalism therefore provides the formal basis for treating the GP collective not merely as a business model but as a genuine higher-level intelligent system with its own Ψ -band, its own G-loop dynamics, and its own developmental trajectory toward expanding *Volume* — the diversity of near-critical G-loops the collective can reliably enter and re-enter across its operational niche.

42. Group-Level G Definition

At the collective level, general intelligence g can be read off the group control loop in the same way as for individuals:

Collective intelligence is an increasing function of (i) time the group spends in its Ψ -band under standardised challenges, and (ii) the diversity of G-loops the group can reliably enter across its operational domains.

More intelligent collectives spend more time in their collective Ψ -band — maintaining productive GP dynamics across their node network — and can access a richer repertoire of near-critical coordination loops without falling out of their collective viability band.

This definition applies equally to dyads, teams, organisations, and — at civilisational scale — to the multipolar GP network as an alternative to the centralised intelligence architectures that treat collective intelligence as mutual-information maximisation under a single dominant objective function.

Part XIII — Key References

43. Primary Sources for the Integrated Framework

The following references are the core empirical and theoretical foundations for the G loop / Griffiths Phase / Trident-G integrated architecture. They are listed by framework layer.

43.1 Griffiths Phase in Brain Networks

Wu, K., Chialvo, D. R., Zhou, C., & Yu, L. (2026). Optimal Griffiths Phase in Heterogeneous Human Brain Networks: Brain Criticality Embracing Stability and Flexibility across Individuals. *Preprint*. arXiv:2512.03409 / ResearchSquare DOI: 10.21203/rs.3.rs-8410190/v1

- The primary empirical source for the GP framework as applied to individual differences in human brain dynamics
- Demonstrates that synergistic interaction between structural network modularity and regional heterogeneity in local excitability generates a biologically viable GP in human brain networks
- Uses Human Connectome Project data (n=990); shows individual position within GP predicts cognitive profile
- Available open access: <https://arxiv.org/abs/2512.03409>

Moretti, P., & Muñoz, M. A. (2013). Griffiths phases and the stretching of criticality in brain networks. *Nature Communications*, 4, 2521.
<https://doi.org/10.1038/ncomms3521>

- Original demonstration that hierarchical-modular architecture of cortical networks replaces a singular critical point with an extended Griffiths phase
- Foundational paper showing GP in both synthetic hierarchical networks and empirical brain networks including the human connectome
- Available open access

Wang, R., Liu, M., Cheng, X., Wu, Y., Hildebrandt, A., & Zhou, C. (2021). Segregation, integration, and balance of large-scale resting brain networks configure different cognitive abilities. *Proceedings of the National Academy of Sciences*, 118(23), e2022288118. <https://doi.org/10.1073/pnas.2022288118>

- Empirical basis for the segregation/integration balance and its relationship to cognitive ability profiles
- Shows stronger FC integration associates with better general cognitive ability; stronger segregation fosters crystallised intelligence and processing speed; memory peaks at the balanced state
- Available open access: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8201916/>

43.2 Zhang-Tang: Entropy/Mutual Information Bifurcating Dynamic

Zhang, X.-Y., & Tang, C. (2025). Heavy-tailed update distributions arise from information-driven self-organization in nonequilibrium learning. *Proceedings of the National Academy of Sciences*, 122(51), e2523012122.
<https://doi.org/10.1073/pnas.2523012122>

- Demonstrates that heavy-tailed power-law update distributions in neural networks arise from a fundamental trade-off between maximum entropy (exploration) and mutual information constraint (exploitation)
 - Shows the power-law exponent $\alpha = \lambda\beta$ remains stable throughout training — the signature of self-organised criticality
 - Directly provides the computational-level mechanism for the FPCN-A/B bifurcating dynamic
 - Available open access: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12745802/>
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43.3 Successor Representation

Stachenfeld, K. L., Botvinick, M. M., & Gershman, S. J. (2017). The hippocampus as a predictive map. *Nature Neuroscience*, 20(11), 1643–1653.
<https://doi.org/10.1038/nn.4650>

- Foundational paper establishing that hippocampal place cells encode successor representations — predictive maps of future states weighted by temporal discounting
 - Argues that entorhinal grid cells encode a low-dimensional basis set for the predictive representation
 - Provides the neural substrate for the intermediate-horizon SR layer of the G loop architecture Gershman, S. J. (2018). The successor representation: its computational logic and neural substrates. *Journal of Neuroscience*, 38(33), 7193–7200. <https://doi.org/10.1523/JNEUROSCI.0151-18.2018>
 - Review of the computational logic and neural evidence for successor representations
 - Frames SR as a solution to the efficiency/flexibility trade-off in reinforcement learning
 - Available open access: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6096039/>
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43.4 Neuronal Avalanches and Criticality

Beggs, J. M., & Plenz, D. (2003). Neuronal avalanches in neocortical circuits. *Journal of Neuroscience*, 23(35), 11167–11177.
<https://doi.org/10.1523/JNEUROSCI.23-35-11167.2003>

- Original demonstration that spontaneous activity propagation in cortical networks follows power-law avalanche statistics
 - Establishes the empirical signature of criticality that the α stability monitor in the G loop tracks
 - Foundational paper for the critical brain hypothesis
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43.5 Free Energy Principle and Active Inference

Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138. <https://doi.org/10.1038/nrn2787>

- Foundational paper for the free energy framework that Trident-G extends
 - Proposes that the brain minimises surprise/prediction error — the computational principle that Trident-G refines into band-maintenance rather than zero-minimisation
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43.6 Trident-G

Smith, M. A. (2025). Trident G: A Computational-Level Theory of Intelligent Life. *University of Essex Online*. December 2025.

- The primary source for the allostatic grounding, cusp geometry, meta-parameter architecture, and Ψ -band formulation integrated throughout Parts VI–XII of this document
 - Proposes that intelligence is the navigable G-volume an organism can traverse while remaining within its Ψ -band
 - Available as preprint from the author (ma18183@essex.ac.uk)
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43.7 Piecewise Learning Curves

Donner, Y., & Hardy, J. L. (2015). Piecewise power laws in individual learning curves. *Psychonomic Bulletin & Review*, 22(5), 1308–1319. <https://doi.org/10.3758/s13423-015-0811-x>

- Empirical analysis of 25,280 individual learning curves across four cognitive tasks

- Shows individual curves often contain multiple power-law segments with transition points where performance briefly drops before a stronger curve emerges
 - Provides the empirical learning curve evidence that the G loop's transition events explain mechanistically
 - Available open access: <https://pmc.ncbi.nlm.nih.gov/articles/PMC4577530/>
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43.8 Additional References from Trident-G Source Document

The following references appear in the original Trident-G paper and support specific components of the integrated framework:

Friston, K., FitzGerald, T., Rigoli, F., Schwartenbeck, P., & Pezzulo, G. (2017). Active inference: A process theory. *Neural Computation*, 29(1), 1–49.

https://doi.org/10.1162/NECO_a_00912

Sterling, P., & Eyer, J. (1988). Allostasis: A new paradigm to explain arousal pathology. In S. Fisher & J. Reason (Eds.), *Handbook of life stress, cognition and health* (pp. 629–649). John Wiley & Sons.

- Foundational allostasis paper; basis for Trident-G's defended band concept
McEwen, B. S., & Stellar, E. (1993). Stress and the individual: Mechanisms leading to disease. *Archives of Internal Medicine*, 153(3), 209–221.
<https://doi.org/10.1001/archinte.153.3.209>

Aston-Jones, G., & Cohen, J. D. (2005). An integrative theory of locus coeruleus–norepinephrine function: Adaptive gain and optimal performance. *Annual Review of Neuroscience*, 28, 403–450.

<https://doi.org/10.1146/annurev.neuro.28.061604.135709>

- Basis for the explore–exploit tilt δ and its neuromodulatory implementation
Barrett, L. F., & Simmons, W. K. (2015). Interoceptive predictions in the brain. *Nature Reviews Neuroscience*, 16(7), 419–429.
<https://doi.org/10.1038/nrn3950>
 - Foundational paper linking interoceptive inference to the allostatic prediction error ϵ
Thom, R. (2018). *Structural stability and morphogenesis: An outline of a general theory of models*. CRC Press.
<https://doi.org/10.1201/9780429493027>
 - Mathematical foundation for the cusp catastrophe geometry used in Trident-G
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43.9 Working Memory Transfer — Negative Evidence

The following papers establish the failure of standard working memory training to produce far transfer, against which the G loop's training architecture is positioned:

Redick, T. S., et al. (2013). No evidence of intelligence improvement after working memory training: A randomized, placebo-controlled study. *Journal of Experimental Psychology: General*, 142(2), 359–379.

Melby-Lervåg, M., Redick, T. S., & Hulme, C. (2016). Working memory training does not improve performance on measures of intelligence or other measures of far transfer. *Perspectives on Psychological Science*, 11(4), 512–534. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC4968033/>